ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a feather	Air resistance matters for feathers.
2	**(d)** slowly stops swinging	Friction at the pivot and air resistance.
3	**(a)** 2500	1 g/cm3 = 1000 kg/m3.
4	**(a)** 8 m/s	100/12.5.
5	**(a)** supports the law	Agreement supports it.
6	**(b)** 20	72 x 5/18.
7	**(c)** 16.7 m/s	60 km/h x 5/18.
8	**(c)** low pressure and high temperature	Real gases approach ideal behaviour.
9	**(a)** 0.15	0.5 x 0.3.
10	**(c)** 0.2	2 x 0.5 x 0.2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acceptable: air hockey puck on a smooth table (1); Fails: car braking on a rough road (1).	1+1
12	Easy to read and compare (1); 3 x 10^8 (1).	1+1
13	Hypothesis: proposed explanation (1); Prediction: expected result from the hypothesis (1).	1+1
14	Sum = 25.5 (1); Mean = 5.1 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	T = 32/20 = 1.6 s (1); Reduces reaction-time error (1); Gives better precision (1).	1+1+1
16	Atom about 10^-10 m (1); Sun about 1.5 x 10^11 m (1); Easy comparison (1).	1+1+1
17	Points follow a trend (1); A stray point stands out (1); Check and repeat (1).	1+1+1
	OR: 0.07 mm = 7 x 10^-5 m (1); = 70 micrometres (1); 1 mm = 1000 micrometres (1).	1+1+1
18	Distance light travels in one year (1); About 9.46 x 10^15 m (1); Used for stars because they are very far (1).	1+1+1
	OR: Wear goggles: protect the eyes (1); Do not taste: poisons (1); Add acid to water slowly: avoid splashes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Periods: 1.0, 1.42, 1.74, 2.0 s (1); T increases with length (1); Increase is not proportional (1); No friction or air resistance assumed (1); Small amplitude assumed (1).	1+1+1+1+1
	OR: Initial observation (1); Hypothesis (1); Repeated experiments (1); Peer review (1); Wide acceptance (1).	1+1+1+1+1
20	Air breathed in a day (1); Heartbeats in a life (1); Mass of water in a school tank (1); Assumed average values (1); Uncertainty from approximations (1).	1+1+1+1+1
	OR: Mix of litres and gallons or kg and pounds (1); Wrong fuel load (1); Safety risks (1); Need for SI units (1); Double-checking (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** defines its limits and develops a better model	Model improvement.
2	**(c)** 12.6 g	Average.
3	**(a)** 2500 cubic cm	2.5 x 1000.
4	**(a)** 'Beauty is the best colour of a flower.'	Cannot be measured.
5	**(a)** 2 per cent	1/50 x 100.
6	**(c)** check the data and refine the model	Honest approach.
7	**(a)** The liquid has a mass of 30 g.	Has a number.
8	**(b)** 3 x 10^5	Divide by 1000.
9	**(b)** 4	1, 2, 4, 3.
10	**(a)** 5.2 cm	Subtract the zero error.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Error = 0.2 cm (1); 0.2/10 x 100 = 2 per cent (1).	1+1
12	Standard convention in science (1); Prevents confusion (1).	1+1
13	(a) 1.5 m (1); (b) 1500 mm (1).	1+1
14	Smallest value measurable by the instrument (1); 1 mm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Initial: heavier ball falls faster (1); Experiment: both fall together (1); Modified: fall depends on air resistance not mass (1).	1+1+1
16	Reaction time at start and stop (1); Repeat measurements (1); Average values (1).	1+1+1
17	Newton's laws: motion (1); Law of conservation of mass: reactions (1); Law of conservation of energy: energy changes (1).	1+1+1
	OR: Accuracy: closeness to true value (1); Precision: closeness of repeated readings (1); A balance with zero error is precise but inaccurate (1).	1+1+1
18	Independent: light (1); Dependent: colour (1); Tomatoes ripen in both; heat and chemical changes drive ripening (1).	1+1+1
	OR: Clear communication (1); Same meaning everywhere (1); Avoids errors (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dalton: indivisible atom (1); Thomson: electrons in a positive sphere (1); Rutherford: nucleus (1); Bohr: shells (1); Each model explained new facts (1).	1+1+1+1+1
	OR: Reports and journals (1); Peer review (1); Honesty prevents false data (1); Humility accepts correction (1); Cooperation across countries like CERN (1).	1+1+1+1+1
20	Accuracy (1); Precision (1); Least count (1); Zero error corrected by subtraction (1); Examples (1).	1+1+1+1+1
	OR: One plot is not enough (1); Need a control plot (1); Same soil and water (1); Repeat on many plots (1); Statistical comparison (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** m/s	Combination of base units.
2	**(b)** an estimate by comparison	Counting units.
3	**(a)** 1 mm	Smallest division.
4	**(b)** one-thousandth	10^-3.
5	**(d)** 31622400	366 x 86400.
6	**(d)** 0.012 cm	1.2/100.
7	**(c)** one-millionth	10^-6.
8	**(d)** 3000	3 x 1000.
9	**(c)** air resistance is negligible	Common idealisation.
10	**(d)** 1000000	10^6.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Confirms results are not due to chance (1); Detects hidden errors (1).	1+1
12	Cannot place it on a balance (1); Use gravitational laws and planetary motion (1).	1+1
13	Cannot answer questions that cannot be tested (1); Results depend on measurement accuracy (1).	1+1
14	Beats per year = 72 x 60 x 24 x 365 = 3.8 x 10^7 (1); In 70 years about 2.6 x 10^9 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Easy to picture (1); Electrons do not move in fixed orbits (1); Later models were developed (1).	1+1+1
16	Observation: dark clouds (1); Hypothesis: clouds bring rain (1); Prediction: rain will fall (1).	1+1+1
17	Estimate roughly the expected size (1); Compare the exact calculation (1); A big difference signals an error (1).	1+1+1
	OR: (a) 30 m/s (1); (b) 54 km/h (1); (c) 5000 kg/m3 (1).	1+1+1
18	System is complex (1); Small differences grow (1); Data are limited (1).	1+1+1
	OR: Evidence can be checked (1); Authorities may be wrong (1); Ideas change with data (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mean = 1.83 mm (1); Error = 0.03 (1); Percentage = 1.67 per cent (1); Readings are precise (1); Slight inaccuracy (1).	1+1+1+1+1
	OR: Number of reliable digits (1); 12.43 s has four (1); Rounding to 12.4 s three figures (1); Final answer cannot be more precise than the data (1); Zeros rules (1).	1+1+1+1+1
20	Observation of falling bodies and planets (1); Mathematical description (1); Predictions tested (1); Agreement supports the law (1); Limits recognised (1).	1+1+1+1+1
	OR: Questions about matter (1); Experiments (1); Atomic models (1); Applications: medicines, batteries (1); Responsibility for the environment (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.